VELS

INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
(Deemed to be University Estd. u/s 3 of the UGC Act, 1956)
PALLAVARAM, THALAMBUR, PERIYAPALAYAM, THIRUVANMIYUR - CHENNAI



# VISION GUARD: YOLOV11-ENABLED AUTONOMOUS THREAT INTELLIGENCE FOR MODERN SECURITY INFRASTRUCTURE


## TEAM MEMBERS

ARNOB SAMANTA (23619103)

M.AKSHATH (23619101)

A.M.PON MADHU (23619128)

## INTERNAL GUIDE

MS. P. MOHANA PRIYA.

# INTRODUCTION

Traditional CCTV surveillance is limited by human fatigue and high false-positive rates. Our project modernizes security by deploying a YOLOv11-based intelligent pipeline. By training on the Roboflow Weapon Detection Dataset, we enable real-time, high-precision threat identification. Through motion-gated inference and temporal buffering, our system autonomously filters environmental noise, extracting high-fidelity data for actionable threat intelligence. This approach drastically minimizes the cognitive burden on human operators while ensuring rapid, reliable response times in critical scenarios. Ultimately, the system provides a robust, scalable framework designed for seamless integration into existing security infrastructure.

# ABSTRACT

This project introduces an innovative, high-performance threat detection and data extraction pipeline designed to modernize public security surveillance. Leveraging the state-of-the-art YOLOv11 architecture, the system is fine-tuned on the Roboflow Weapon Detection Dataset, a comprehensive, consolidated collection of over 20,000 annotated images of firearms and knives, to ensure real-time, high-precision identification. Our implementation features a robust pre-processing engine that utilizes motion-gated inference via background subtraction to optimize computational efficiency and Contrast-Limited Adaptive Histogram Equalization (CLAHE) to maintain reliability in challenging low-light conditions. To eliminate transient noise and flickering, we employ a temporal buffering mechanism that mandates consistent detection across consecutive frames. Upon verification of a threat, the system dynamically extracts the Region of Interest (ROI), precisely isolating the subject's face and posture to generate high-fidelity data suitable for behavioral analysis. By transitioning from basic object classification to a sophisticated, context-aware framework, this project bridges the critical gap between raw video input and actionable intelligence, enabling security infrastructure to autonomously filter irrelevant activity and focus exclusively on verified threats, thereby providing a scalable, robust, and effective solution for the complexities of modern security environments.

# PROBLEM STATEMENT



- Existing  CCTV System monitoring mainly depends on human observation.
- Watching many cameras can be difficult and time-consuming.
- Threats may be detected late.
- Poor lighting, blur, and blocked views can reduce detection accuracy.
- An automated real-time system is needed to detect threats and send quick alerts.

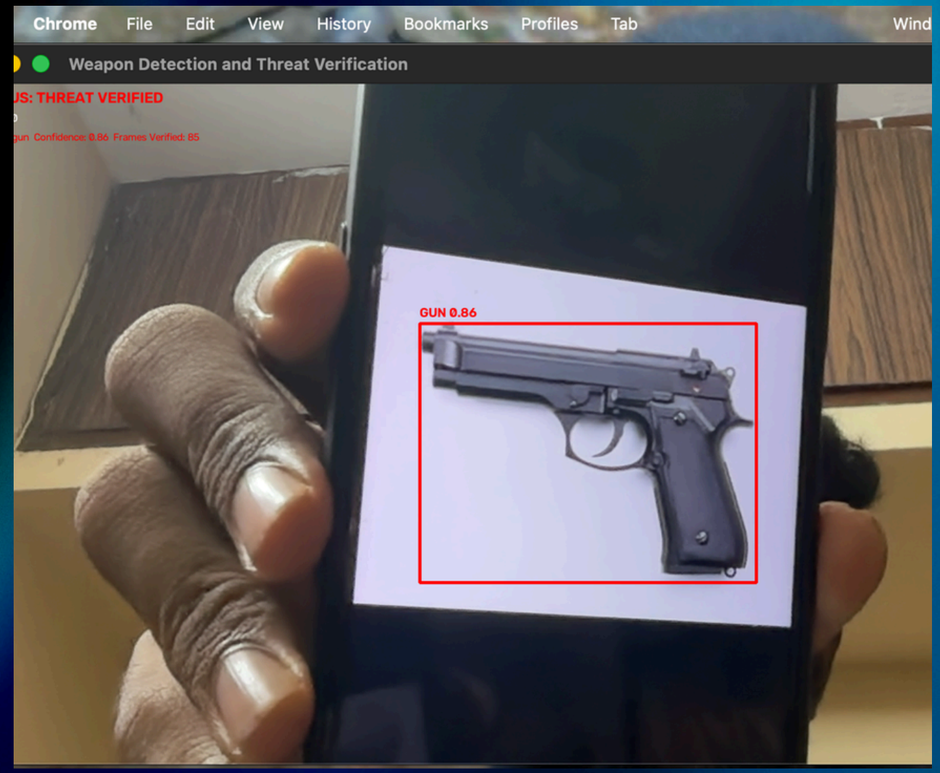
# OBJECTIVE




- Accurate Detection: Train YOLOv11 using 20,000+ weapon images.
- Faster Processing: Use motion detection to process only important frames.
- Better Detection: Apply CLAHE to improve detection in low-light conditions.
- Noise Reduction: Use temporal buffering to reduce false and flickering detections.
- ROI Extraction: Focus on important areas for further behavior analysis.

# METHODOLOGY

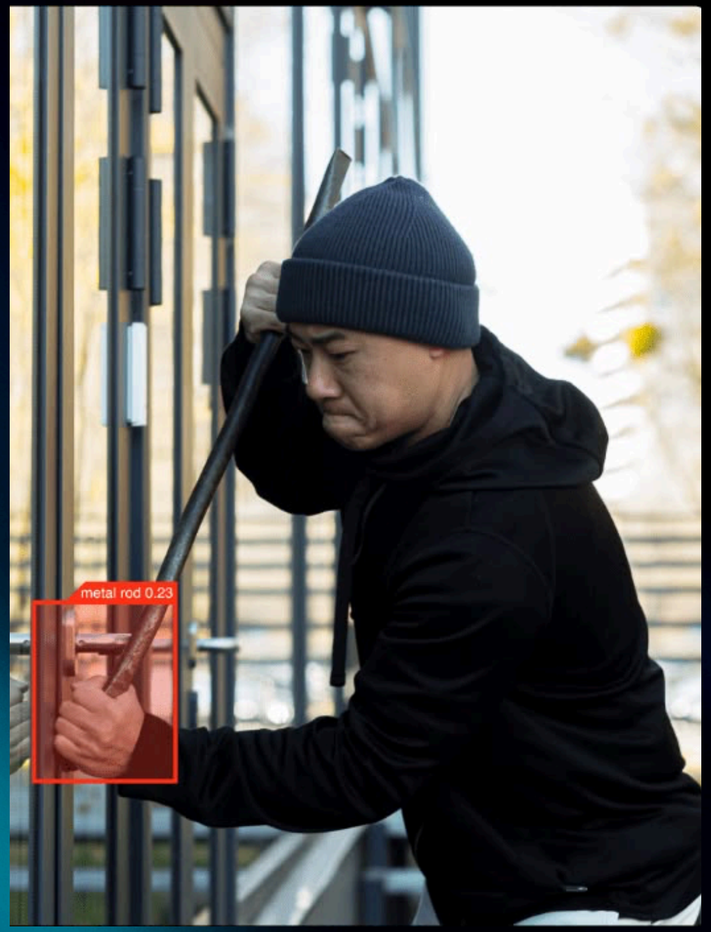
- Live CCTV monitoring
- YOLOv11 weapon/object detection
- Confidence-based false-alert reduction
- Low/Medium/High threat classification
- Instant alerts with location, time, confidence & image
- Incident storage for investigation



# LITERATURE SURVEY

| Ref # | Authors, Year, Title /Domain Focus | Key Methodology | Limitation / Research Gap |
|---|---|---|---|
| 1 | Chen et al. (2024). Static Object Detection for Firearms in CCTV | Standard YOLOv8 trained on small-scale custom datasets. | Suffers from massive false positives in ordinary environments (e.g., kitchens, desks); lacks context. |
| 2 | Kumar & Sharma (2023). Real-Time Surveillance and Motion Tracking | Background subtraction combined with basic bounding-box alarms. | Ignores environmental lighting shifts (night/dim areas) and lacks pre-processing enhancement. |
| 3 | Zhang et al. (2025). Deep Learning for Weapon Classification | Convolutional Neural Networks (CNNs) focused heavily on precision metrics. | High computational lag; fails to handle partial occlusions or small, distant objects in wide CCTV angles. |
| 4 | Rodriguez et al. (2023). Smart Alert Systems in Public Infrastructure | Instant push-notification triggers based on single-frame triggers. | Prone to alert fatigue security staff due to lack of temporal consistency checks (flickering false alarms). |
| 5 | Patel & Gomez (2024). Behavioral Pre-requisites in Threat Detection | Multi-stage pipeline dividing object localization from intent analysis. | Often drops context during hand-off between detection and downstream analysis modules. |

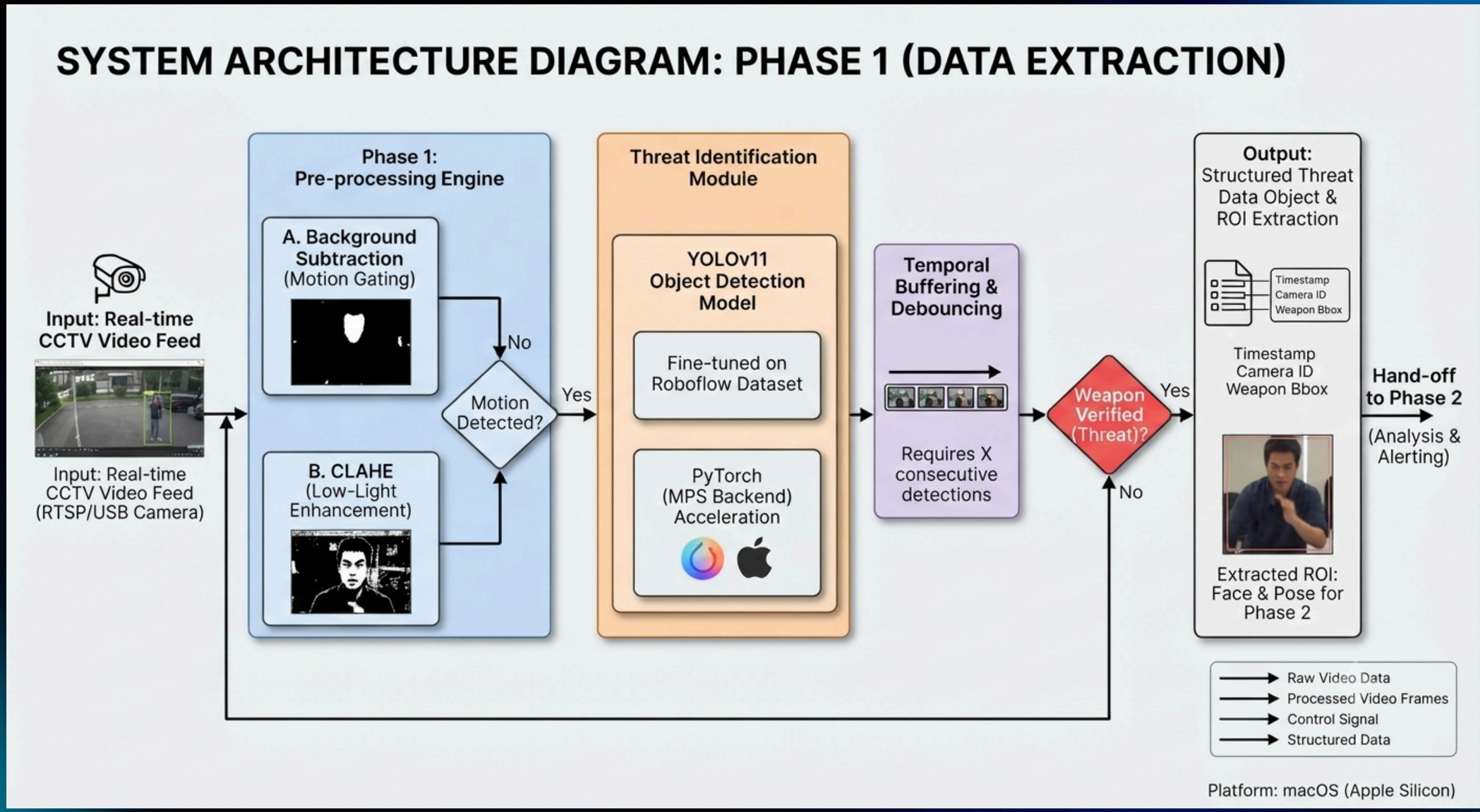
# RESEARCH GAP




Existing CCTV-based threat detection systems mainly focus on object detection and may not perform well in real-world conditions. Challenges such as poor lighting, blur, occlusion, unnecessary frame processing, and unstable detections can reduce accuracy and increase false alerts. Most existing approaches also lack a unified method for motion-based processing, image enhancement, temporal filtering, and ROI extraction. Therefore, there is a need for a lightweight and reliable real-time surveillance system that can handle these challenges and provide accurate threat detection with faster alerts.

| Software Components & Specifications | Hardware Components & Specifications |
| --- | --- |
| Operating System: macOS Sonoma (14.x) or macOS Sequoia (15.x) | Processor / Architecture: Apple Silicon Mac (M1 Pro/Max, M2, M3, or M4 series chip) with Unified Memory architecture |
| Programming Language: Python 3.10+ (managed via Miniconda or Homebrew) | Memory (RAM): Minimum 16 GB Unified Memory (for smooth local model inference and processing) |
| Deep Learning Framework: PyTorch (utilizing native Apple Silicon MPS - Metal Performance Shaders GPU acceleration) | Storage: 50 GB+ available SSD space (for datasets, virtual environments, and model checkpoints) |
| Detection Model: Ultralytics YOLOv11 (fine-tuned on Roboflow weapon dataset) | Camera Input: Continuity Camera (using iPhone) or external USB Webcams / simulated RTSP streams |

# SYSTEM ARCHITECTURE



# GANTT CHART

## GANTT CHART – YOLOv11 BASED THREAT DETECTION AND DATA EXTRACTION PIPELINE

TODAY: 18/08/2026 (AUGUST WEEK 3)

| No. | Project Activity | JULY 2026 | | | | AUGUST 2026 | | | | SEPTEMBER 2026 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | WEEK 1 | WEEK 2 | WEEK 3 | WEEK 4 | WEEK 1 | WEEK 2 | WEEK 3 | WEEK 4 | WEEK 1 | WEEK 2 | WEEK 3 | WEEK 4 |
| 1. | Literature Review & Problem Definition | ■ | ■ | ■ | ■ | | | | | | | | |
| 2. | System Architecture & Planning | | ■ | ■ | ■ | | | | | | | | |
| 3. | Roboflow Weapon Dataset Collection | | | ■ | ■ | ■ | | | | | | | |
| 4. | Dataset Cleaning & Pre-processing | | | | ■ | ■ | ■ | ■ | | | | | |
| 5. | YOLOv11 Model Configuration | | | | | ■ | ■ | ■ | | | | | |
| 6. | YOLOv11 Model Training & Fine-tuning | | | | | | ■ | ■ | ■ | | | | |
| 7. | Model Validation & Performance Evaluation | | | | | | ■ | ■ | ■ | ■ | | | |
| 8. | Motion-Gated Inference / Background Subtraction | | | | | | ■ | ■ | ■ | ■ | | | |
| 9. | CLAHE Low-Light Enhancement | | | | | | ■ | ■ | ■ | ■ | | | |
| 10. | Temporal Buffering & Consistent Detection | | | | | | | ■ | ■ | ■ | | | |
| 11. | Complete Pipeline Integration | | | | | | | | | ■ | ■ | | |
| 12. | Real-time Testing & Optimization | | | | | | | | | | ■ | ■ | |
| 13. | Accuracy / Precision / Recall Evaluation | | | | | | | | | | | ■ | ■ |
| 14. | Final Documentation & Report | | | | | | | | | | | ■ | ■ |
| 15. | Final Demo & Project Presentation | | | | | | | | | | | | ■ |

**LEGEND**

- Completed
- In Progress (as of 18/08/2026)
- Planned / Upcoming

**STATUS AS OF 18/08/2026 (AUGUST WEEK 3)**

**COMPLETED**

- Literature Review & Problem Definition
- System Architecture & Planning
- Dataset Collection
- Dataset Cleaning & Pre-processing
- YOLOv11 Model Configuration
- Initial Model Training & Fine-tuning
- Initial Model Validation

**IN PROGRESS**

- Model Training & Optimization
- Motion-Gated Inference
- CLAHE Enhancement
- Temporal Buffering Mechanism
- Pipeline Integration (Initial Phase)

**PLANNED (NEXT)**

- Complete Pipeline Integration
- Real-time Testing & Optimization
- Accuracy / Precision / Recall Evaluation
- Final Documentation & Report
- Final Demo & Project Presentation

# CONCLUSION



IN CONCLUSION, THIS PROJECT PRESENTS A RELIABLE AND INTELLIGENT APPROACH TO MODERN SECURITY SURVEILLANCE. BY INTEGRATING YOLOV11-BASED WEAPON DETECTION WITH MOTION-GATED INFERENCE, LOW-LIGHT IMAGE ENHANCEMENT, AND TEMPORAL BUFFERING, THE SYSTEM ACCURATELY IDENTIFIES VERIFIED THREATS WHILE REDUCING FALSE ALARMS CAUSED BY ENVIRONMENTAL NOISE. THIS ENABLES SECURITY PERSONNEL TO FOCUS ON CRITICAL EVENTS, RESPOND MORE QUICKLY, AND MAKE BETTER-INFORMED DECISIONS. WITH ITS SCALABLE ARCHITECTURE AND COMPATIBILITY WITH EXISTING CCTV INFRASTRUCTURE, THE PROPOSED SYSTEM OFFERS A PRACTICAL STEP TOWARD SAFER, SMARTER, AND MORE EFFICIENT PUBLIC-SECURITY MONITORING.

# THANK YOU